

Causal Site Maps Can Fail to Replicate While Residual Directions Transfer

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Abstract

Activation patching is commonly summarized as a layer-by-position map: sites are scored individually, then the resulting heatmap is interpreted as the causal footprint of a behavior. We test whether replication of that map is necessary for a reusable intervention. In Qwen2.5-7B-Instruct, full patching maps for two elicited micro-skills—Variable substitution and explicit Completion state—fail a pre-registered template-disjoint replication gate ($\rho = 0.37$ and 0.14). For Variable, however, the expected-position column is stable across templates ($r = 0.999$), and a mean counterfactual-minus-clean residual direction extracted at that site transfers to held-out templates at 1.01 times within-template strength. The same intervention test transfers across Completion surfaces (1.00; 100% greedy flips), at 1.5B for Variable (1.02), and in both directions across a 13-token position shift (1.09 and 0.92). An embedding control fails and the negative direction hurts, but an unrelated-value direction also transfers (0.59; cosine 0.77 with the matched direction), producing our pre-registered **GENERIC_BOOST** verdict. The supported object is therefore a coarse, signed, layer-computed *slot-update intervention*, not a value-specific binder. Our result is a controlled counterexample to “non-replicating site-map implies no reusable intervention.” It does not establish that the added direction is the model’s unique natural mechanism.

1 Introduction

Activation patching, causal tracing, and interchange interventions ask whether replacing an internal state changes model behavior [8, 13, 20]. Sweeping the intervention over layers and token positions yields an intuitive heatmap of *where* a behavior is causally sensitive. Such maps are useful, but their visual sharpness can encourage a stronger inference: if the map does not replicate across surface forms, the mechanism itself is fragile.

That inference confounds two objects. A site-map scores scalar effects indexed by coordinates. A direction test asks whether the *activation displacement* learned at one set of sites can be reused at another. These need not have the same invariances.

We make this distinction concrete in controlled minimal-pair tasks. We first calibrate a standard patching harness on entity–attribute binding. We then map two elicited micro-skills at 7B and apply a pre-registered template-replication gate. Both maps fail. We preserve those failures, then test a separately pre-registered hypothesis: the reusable object may be a residual direction at a prompt-conditioned site rather than the complete layer-by-position map.

The direction transfers at approximately within-template strength across held-out surfaces for both skills. For Variable, it also transfers at 1.5B and between positions separated by an inert 13-token prefix. Controls narrow the claim: the effect is computed beyond token embeddings and reverses under negative addition, but an unrelated-value direction retains 59% of matched strength. We therefore identify a coarse value-slot update, not lexical binding.

Contribution. Difference-of-means steering, task vectors, function vectors, and large cross-template transfer studies are established [9, 14, 16, 18, 19]. Our contribution is not another discovery of directions. It is the paired, pre-registered observation that a site-map replication gate can reject the same behavior for which a held-out direction-transfer gate succeeds. This supplies a simple methodological warning: map fragility is not sufficient evidence of intervention fragility.

2 Related work and the 2026 reliability context

Patching is method-dependent. Activation-patching results depend on corruption, intervention direction, and metric [8, 22]. Causal scrubbing and related approaches emphasize testing explicit invariances rather than reading a

single intervention as a complete explanation [3, 10]. Our template-disjoint gate follows that spirit, but evaluates a simpler layer-by-position object. Recent work also finds substantial variance under paraphrase and resampling [12], prompt-specific circuit structure [5], and different circuit structures that can implement interchangeable functions [1]. Site fragility is therefore neither new nor universal.

Activation directions. Task and function vectors can compactly mediate in-context functions [9, 18]; activation addition and contrastive activation addition steer behavior using contrastive residual directions [16, 19, 23]. Generalization is uneven across prompts and datasets [2, 17]. Nadaf [14] provide the closest large-scale precedent for cross-template function-vector transfer; Cheng and Zhang [4] instead find that some in-context functions require coordinated multi-position interventions. Most directly, Opielka et al. [15] show that causal function vectors need not be format-invariant and distinguish them from more invariant concept vectors. Our result is complementary: within a narrow controlled format, the full site-map is less invariant than the intervention direction extracted at its stable site. Related dissociations between causal localization and successful manipulation have also appeared in knowledge editing [7].

Faithfulness warning. Successful intervention is not by itself a faithful mechanistic explanation. Subspace patching can recruit dormant pathways [11], and causal interventions can move representations away from the model’s natural distribution [6]. Our controls reject several simpler explanations but do not measure representational divergence. We consequently claim a transferable *intervention direction*, not that the model’s unique natural computation lives in that one-dimensional subspace.

3 Experimental setup

3.1 Tasks and elicitation

We use Qwen2.5-Instruct [21] at 7B (8-bit weights, fp16 residual interventions) and 1.5B (unquantized), with greedy decoding and chat-templated prompts. Minimal pairs change one computational variable and have single-token answer alternatives.

- **Binding calibration:** two entity–attribute assignments; the counterfactual swaps an answer-relevant attribute.
- **Variable substitution:** “Let $X = \text{dog}$ ” versus “Let $X = \text{cat}$ ”, followed by a query for X . Five variable-name surfaces have ten value pairs each.
- **Completion state:** an explicit two-rule table maps a Boolean state to one of two actions. Four surface tags have ten instances each.

Variable and explicit Completion reach 100% behavioral accuracy in the retained sets. An implicit Completion variant without the rule table reaches 0% and is excluded. An Instruction-versus-mention skill is dropped after two elicitation failures. These exclusions were decided by behavioral gates before patching.

3.2 Patching and map replication

For clean input x and counterfactual x' , let $h_{\ell,p}$ denote the post-layer residual at layer ℓ , position p . We patch $h_{\ell,p}(x) \leftarrow h_{\ell,p}(x')$ and measure

$$\text{IE}_{\ell,p} = d(x; h_{\ell,p} \leftarrow h_{\ell,p}(x')) - d(x),$$

where $d = \text{logit}(a_{\text{cf}}) - \text{logit}(a_{\text{clean}})$ at the final position. Logit difference is continuous and approximately linear in the residual stream, as recommended for patching studies [8].

The pre-registered P1 gate computes maps on template-disjoint halves and requires Spearman $\rho > 0.5$ and superiority to a pair-partition null at $p < 0.01$. A random-position null separately tests the expected site.

3.3 Direction transfer

At a frozen layer and slot, donor templates define

$$\Delta = \frac{1}{n} \sum_i (h_{\ell,p}(x'_i) - h_{\ell,p}(x_i)).$$

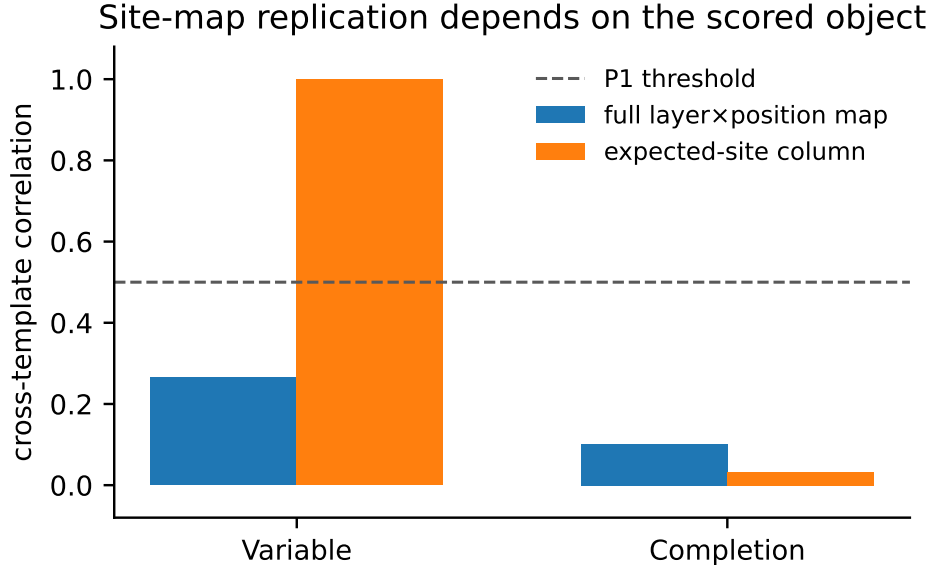


Figure 1: **The scored object changes the replication verdict.** Pairwise correlations across Variable and Completion surfaces. Variable’s full maps are fragile while its value-position column is nearly identical across surfaces. The dashed line is the pre-registered P1 threshold.

We add $\alpha\Delta$ to clean held-out targets. A target cell passes when (i) mean ΔIE is positive, (ii) it beats 100 same-norm random directions at $p < 0.01$, and (iii) cross-template ΔIE divided by a direction learned within that target is at least 0.5. Primary tests use $\alpha = 1$ and layer 2, frozen from the 7B maps.

4 Results

4.1 The instrument recovers a known causal site

On 30 entity–attribute Binding pairs at 7B, patching the queried entity’s attribute position moves answer log-odds in the predicted direction against a matched random-position null ($p = 0.001$). The effect replicates on two disjoint 15-pair halves (both $p = 0.001$), and the top site is the known changed attribute. This calibration blocks the explanation that downstream failures are caused by a nonfunctional harness.

4.2 Sharp site-maps fail template replication

Both novel skills fail P1. Variable has a large expected-site effect ($\text{IE} = 53.3$, $p = 0.001$) and a sharp top site at layer 2, position 27, yet its template-half map correlation is only $\rho = 0.37$ ($p = 0.28$). Completion has $\rho = 0.14$; its top effect is at position 62, one token after the pre-registered bit at position 61.

The Variable failure is specifically about the full map. Across all ten template pairs, mean full-grid correlation is 0.26, while the expected-position column alone correlates at 0.999 (Figure 1). A stable causal column sits inside non-replicating off-site geometry. “Map failure” here always means failure under this frozen full-grid scoring rule; it does not imply that every semantically aligned site description is unstable.

4.3 Directions transfer where maps do not

For Variable at 7B, a direction from surfaces X, Y transfers to held-out Z, W, K with ratios 1.04, 1.14, 0.83 (mean 1.01), each beating its same-norm random-direction null. Leave-one-surface-out transfer has mean ratio 1.07.

The result generalizes in three pre-registered ways (Figure 2). First, Completion direction transfer from surfaces A, B to C, D at the map’s top position yields mean ratio 1.00 and flips the greedy answer on every target. The bit position itself has zero direction norm, consistent with the map peak occurring one token later. Second, Variable at 1.5B passes all three held-out surfaces at layer 2 with mean ratio 1.02. Third, an inert prefix moves the Variable

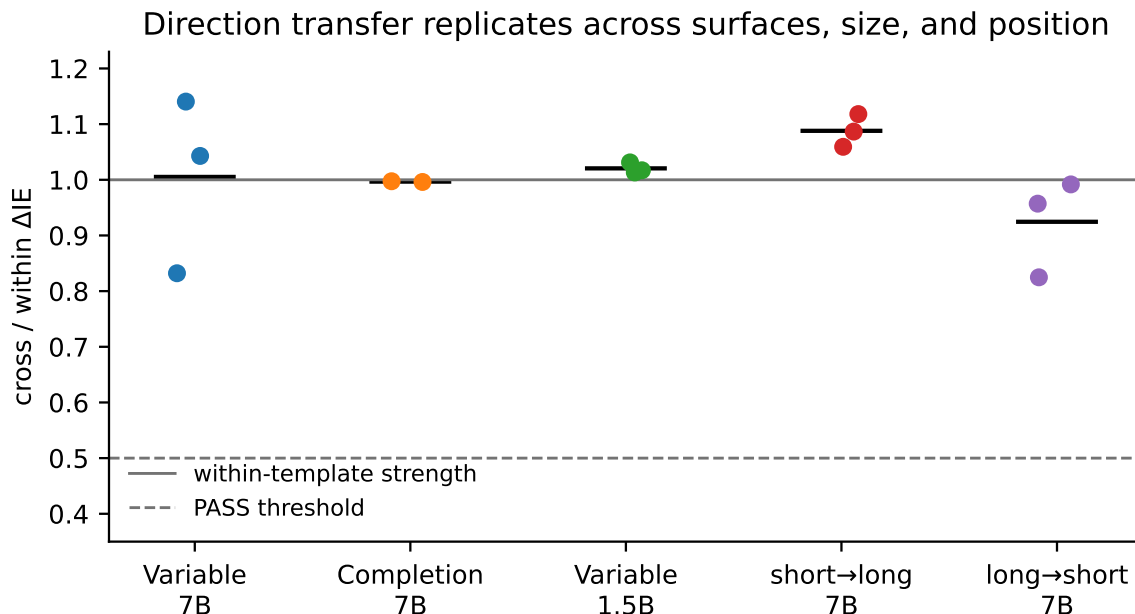


Figure 2: **Held-out transfer relative to a within-surface direction.** Points are target surfaces; black ticks are means. Every primary cell clears the pre-registered 0.5 ratio and random-direction significance gates. Completion was tested at 7B; the Variable protocol was replicated separately at both model sizes.

slot from position 27 to 40; short-to-long and long-to-short directions both pass all targets, with mean ratios 1.09 and 0.92.

4.4 Controls identify a coarse slot update

At 7B, the Variable result is stable for $\alpha = 1, 2$ and for layers 1–4 at $\alpha = 1$; $\alpha = 0.5$ fails the all-target gate. Greedy flips begin only at $\alpha \geq 3$, which we treat as strong steering rather than mechanism evidence.

An input-embedding difference added at the layer-0 output passes zero of three targets and has only 0.005 of the layer-2 cross effect. Thus the useful direction is computed beyond the token embedding. Adding the negative matched direction produces mean $\Delta IE = -0.79$, establishing sign sensitivity.

The decisive caveat is a re-forwarded wrong-value control. We replace each donor’s counterfactual with an unrelated value, recompute the layer-2 direction, and test held-out transfer. It still passes all targets at mean ratio 0.59 and has cosine 0.77 with the matched direction (Figure 3), triggering the pre-registered `GENERIC_BOOST` verdict. The matched vector therefore contains a large generic component. The exact supported interpretation is a coarse “update this value slot” intervention, not a direction carrying a specific lexical binding.

5 Discussion

A coordinate-system result. The conventional map and the direction answer different questions. P1 asks whether the complete scalar effect geometry over sites survives a template change. Direction transfer asks whether a displacement extracted at a selected site can be reused elsewhere. In our tasks, the first fails and the second passes at roughly within-surface strength. Therefore, rejecting the map is not enough to reject reusable activation-space interventions.

What “position-free” means here. The Variable direction transfers between positions 27 and 40 after an inert prefix. This rules out dependence on one absolute token index. It does not show invariance to arbitrary instruction format, language, tokenizer, or model family. This distinction is important given evidence that causally effective function vectors can remain format-specific [15].

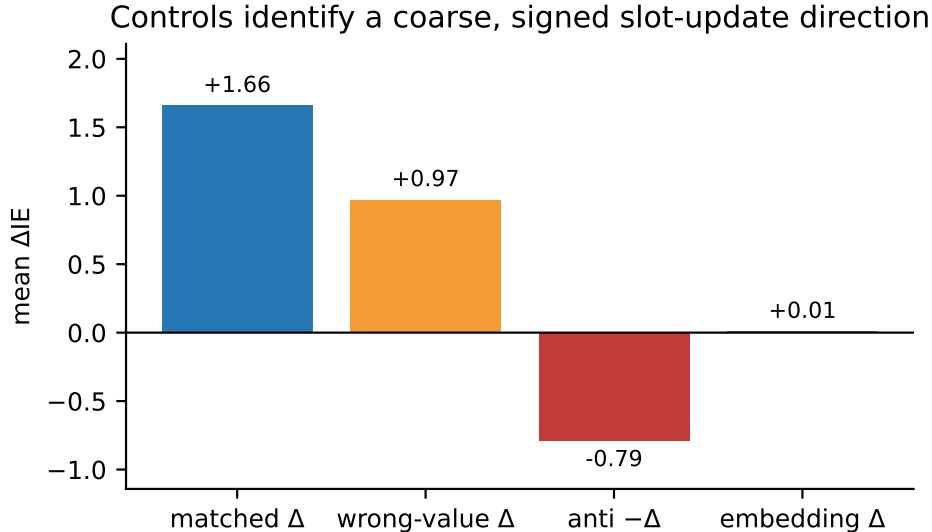


Figure 3: **Controls set the claim strength.** Mean held-out ΔIE for the matched, unrelated-value, negative, and embedding directions. The wrong-value effect is real but weaker; the negative direction hurts; the embedding difference is negligible.

What remains mechanistically unresolved. Our controls show a layer-computed, signed effect and reject equal-norm random directions. They do not prove that activation addition follows the model’s natural computational path. In particular, we do not measure whether $h + \Delta$ remains on the natural latent distribution [6], nor do we rule out dormant-pathway recruitment [11]. “Transferable intervention direction” is therefore the strongest justified term.

Future test. The wrong-value result suggests a decomposition. Build a wrong-value subspace from several unrelated values, project the matched direction onto it, and test the orthogonal residual:

$$\Delta_{\text{specific}} = \Delta_{\text{matched}} - \Pi_{\mathcal{S}_{\text{wrong}}} \Delta_{\text{matched}}.$$

A pre-registered value-selectivity test would ask whether this residual moves the intended counterfactual token more than unrelated values. If successful, the mechanism would decompose into a generic slot update and content-carrying component. A later question is whether independently validated micro-skill directions compose.

6 Limitations

The tasks are synthetic prompt-level micro-skills in one model family. Completion requires an explicit rule table and was tested only at 7B; Variable was tested at 1.5B and 7B. The 7B weights are 8-bit quantized. Direction addition and patching are evaluated primarily through answer logit difference; Variable at $\alpha \leq 2$ does not flip greedy answers. Template and position changes are narrower than format or language shifts. The wrong-value control passes (GENERIC_BOOST). The 1.5B result is same-protocol replication in a different residual space, not cross-model vector transfer. P1 reports failure of a frozen threshold on a small number of surfaces, not an equivalence test proving that all maps differ; the direction ratios likewise are descriptive, not formal equivalence bounds. With 100 random directions, primary control p -values are resolution-limited at $1/101 \approx 0.0099$. The causal site was selected before direction transfer, but target surfaces are not independent of the earlier mapping dataset. Finally, this work localizes sites and interventions, not a complete circuit, and lacks a natural-manifold or dormant-pathway faithfulness test.

7 Conclusion

Two activation-patching maps failed a pre-registered template-replication gate. At their causal peak sites, however, residual directions transferred to held-out surfaces at approximately within-template strength; the Variable protocol

separately replicated at 1.5B and transferred across a 13-token position shift at 7B. The effect is signed and computed beyond token embeddings, but coarse rather than value-specific. The methodological conclusion is narrow and actionable: **a fragile site-map does not imply the absence of a reusable causal intervention direction**. Patching studies that use map replication to judge mechanism reuse should test direction transfer as a separate object.

References

- [1] Alireza Bayat Makou, Jingcheng Niu, Subhabrata Dutta, and Iryna Gurevych. Many circuits, one mechanism: Input variation and evaluation granularity in circuit discovery. *arXiv preprint arXiv:2606.06267*, 2026. URL <https://arxiv.org/abs/2606.06267>.
- [2] Joschka Braun, Carsten Eickhoff, David Krueger, Seyed Ali Bahrainian, and Dmitrii Krashennnikov. Understanding (un)reliability of steering vectors in language models. In *ICLR 2025 Workshop on Foundation Models in the Wild*, 2025. URL <https://openreview.net/forum?id=jf472MdwSP>.
- [3] Lawrence Chan, Adrià Garriga-Alonso, Nicholas Goldowsky-Dill, Ryan Greenblatt, Jenny Nitishinskaya, Ansh Radhakrishnan, Buck Shlegeris, and Nate Thomas. Causal scrubbing: A method for rigorously testing interpretability hypotheses. *AI Alignment Forum*, 2022. URL <https://www.alignmentforum.org/posts/JvZhhzycHu2Yd57RN/causal-scrubbing-a-method-for-rigorously-testing-interpretability>.
- [4] Bryan Cheng and Jasper Zhang. Single-position intervention fails: Distributed output templates drive in-context learning. *arXiv preprint arXiv:2605.04061*, 2026. URL <https://arxiv.org/abs/2605.04061>.
- [5] Gabriel Franco, Lucas M. Tassis, Azalea Rohr, and Mark Crovella. Finding interpretable prompt-specific circuits in language models. *arXiv preprint arXiv:2602.13483*, 2026. URL <https://arxiv.org/abs/2602.13483>.
- [6] Satchel Grant, Simon Jerome Han, Alexa R. Tartaglini, and Christopher Potts. Addressing divergent representations from causal interventions on neural networks. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=cZrTMqYVL6>.
- [7] Peter Hase, Mohit Bansal, Been Kim, and Asma Ghandeharioun. Does localization inform editing? surprising differences in causality-based localization vs. knowledge editing in language models. In *Advances in Neural Information Processing Systems*, 2023. URL <https://openreview.net/forum?id=ElDbUlZtbd>.
- [8] Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024. URL <https://arxiv.org/abs/2404.15255>.
- [9] Roei Hendel, Mor Geva, and Amir Globerson. In-context learning creates task vectors. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9318–9333, 2023. doi: 10.18653/v1/2023.findings-emnlp.624. URL <https://aclanthology.org/2023.findings-emnlp.624/>.
- [10] Yanan Long. Triangulation as an acceptance rule for multilingual mechanistic interpretability. *arXiv preprint arXiv:2512.24842*, 2025. URL <https://arxiv.org/abs/2512.24842>.
- [11] Aleksandar Makelov, Georg Lange, Atticus Geiger, and Neel Nanda. Is this the subspace you are looking for? an interpretability illusion for subspace activation patching. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Ebt7JgMHv1>.
- [12] Maxime M  loux, Fran  ois Portet, and Maxime Peyrard. Mechanistic interpretability as statistical estimation: A variance analysis. *arXiv preprint arXiv:2510.00845*, 2025. URL <https://arxiv.org/abs/2510.00845>.
- [13] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL <https://arxiv.org/abs/2202.05262>.
- [14] Mohammed Suhail B. Nadaf. Steerable but not decodable: Function vectors operate beyond the logit lens. *arXiv preprint arXiv:2604.02608*, 2026. URL <https://arxiv.org/abs/2604.02608>.
- [15] Gustaw Opielka, Hannes Rosenbusch, and Claire E. Stevenson. Causality $\neq$ invariance: Function and concept vectors in LLMs. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=LmLmh6GEL>.

- [16] Nina Rimskey, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 15504–15522, 2024. doi: 10.18653/v1/2024.acl-long.828. URL <https://aclanthology.org/2024.acl-long.828/>.
- [17] Daniel Tan, David Chanin, Aengus Lynch, Brooks Paige, Dimitrios Kanoulas, Adrià Garriga-Alonso, and Robert Kirk. Analysing the generalisation and reliability of steering vectors. In *Advances in Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=v8X70gTodR>.
- [18] Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. Function vectors in large language models. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=AwxytyMwaG>.
- [19] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David S. Udell, Juan J. Vazquez, Ulisse Mini, and Monte Stuart MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023. URL <https://arxiv.org/abs/2308.10248>.
- [20] Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2211.00593>.
- [21] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. URL <https://arxiv.org/abs/2412.15115>.
- [22] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2309.16042>.
- [23] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xu Wang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL <https://arxiv.org/abs/2310.01405>.